Multilevel Methods on Finite Elements

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Modeling subsurface flow

The Idea Implement Multilevel Monte Carlo for subsurface flow via random coefficients. First stage was to implement a multilevel solver in NGSolve that could handle variable coefficients.

We use H_1-conforming piecewise linear finite elements applied to the Poisson problem with variable coefficients and Dirichlet boundary conditions on the unit square.

$$
\begin{aligned}
K(x,y) &= 1 + x^2 + y^2 \\
\sigma &= 0.5 \\
K(x,y)\,\Delta u + \sigma^2 u &= 0 \\
\Omega &= [0,1] \times [0,1]
\end{aligned}
$$

On the left side, $u(0,y) = 1$, on the rest of the boundary $u(x,y)_{\partial\Omega} = 0$.

NGSolve

Netgen/NGSolve is a high performance multiphysics finite element software. It is widely used to analyze models from solid mechanics, fluid dynamics and electromagnetics. Due to its flexible Python interface new physical equations and solution algorithms can be implemented easily.

All in One

Seamless integration from geometric modeling, mesh generation, numerical simulation to visualization

Flexible

Mathematical description of variational formulation allows coupling of arbitrary physical models

Accurate

Cutting edge numerical techniques: high order, vectorial, mixed and discontinuous, Galerkin methods

Efficient

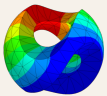
Robust preconditioners adapted to function spaces and differential equations

High Performance

Parallel compute core written in modern C++ combined with flexible steering by Python scripting

Open

Open source based on the LGPL license, extendable by C++ modules and Python packages



Really Big Problems

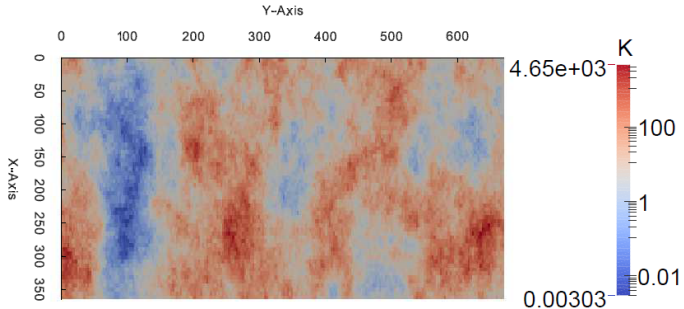
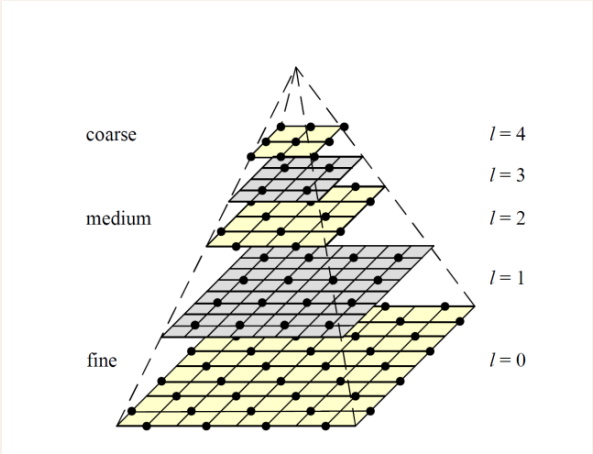


FIG. 2. *SPE10 x-permeability for the top layer.*

TABLE 2
Degrees of freedom, number of nonzeros, and complexities.

Problem	#elements	#faces	#DoFs	nnz	Arithmetic complexity	Operator complexity
60 × 220 (fine)	13200	53080	92680	607485	-	-
30 × 110	3317	13418	23371	152965	1.25252	1.2518
15 × 55	842	3446	5986	39327	1.06491	1.06474
6 × 22	149	634	1093	7351	1.01199	1.0121

A Hierarchy of Finite Element Spaces

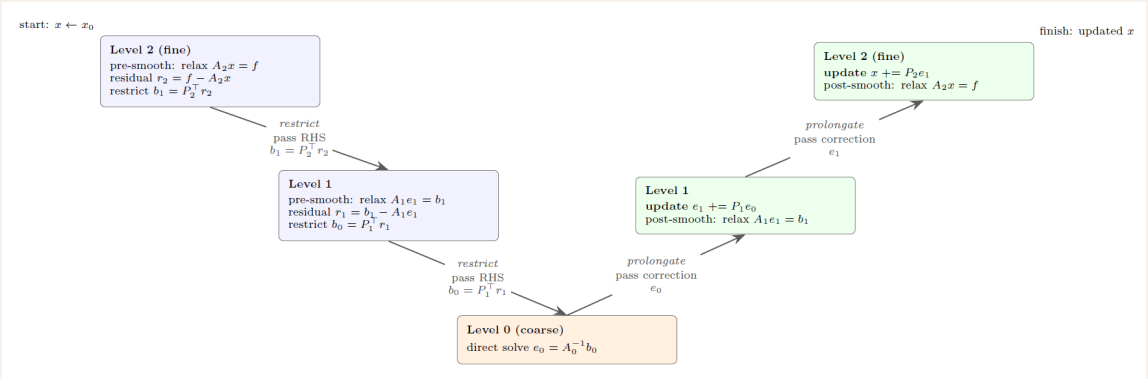


Setting up the finite element spaces

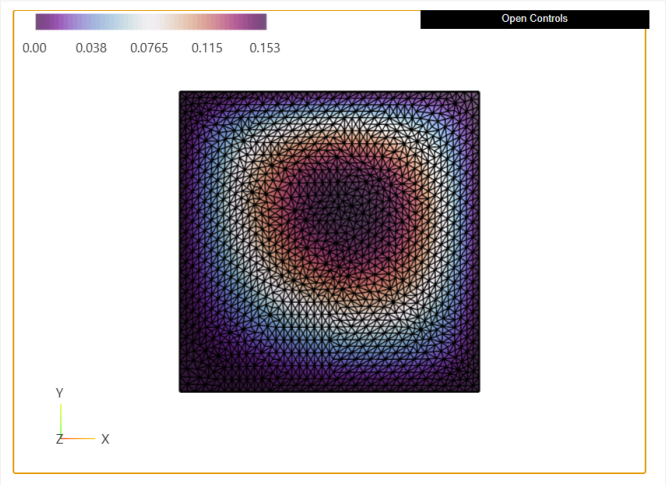
At each level, we have the data storage, solving tools, and projection matrices available.

```
186     mesh: ng.Mesh
187     fes: object
188     a: BilinearForm
189     f: LinearForm
190     gfu: GridFunction
191     free_ids: np.ndarray
192     fixed_ids: np.ndarray
193     P: Optional[object] = None
194     PT: Optional[object] = None
195
```

Solving V-cycle with smoothing and coarse solve

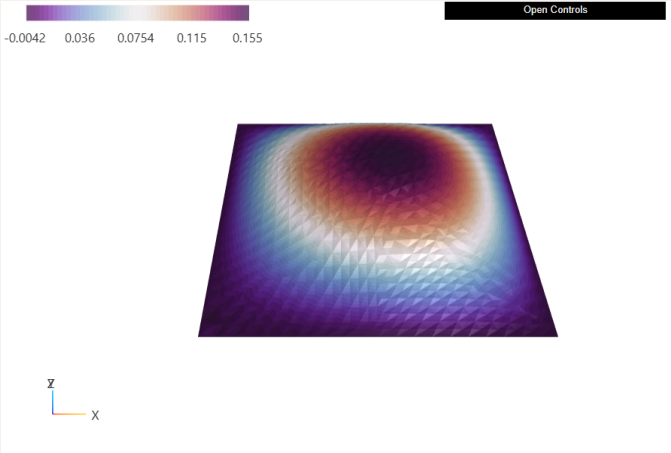


Initial Iterate



fine ndof=1961, coarse ndof=511, P shape 1961x511

Gauss Seidel Smoothing



$\|e\|_E$ (after section 1 pre-smooth): 1.790578

$\|e\|$ L2 error: 9.820784e-04

Example with more layers

lev	ndof	A	P(c->f)	nfree	nfixed	
0	339	339x339	-	305	34	(coarse)
1	1289	1289x1289	1289x339	1223	66	
2	5025	5025x5025	5025x1289	4895	130	
3	19841	19841x19841	19841x5025	19583	258	
4	78849	78849x78849	78849x19841	78335	514	
5	314369	314369x314369	314369x78849	313343	1026	(fine)

Compare to CG

Success! We made it so that we run PCG and when it asks for a preconditioner, it gets our two-level method

PCG (NGSolve CG + V-cycle): 1684 iterations

In contrast, we compared Conjugate Gradient on meshes of different sizes to see how the convergence varies.

Total DoFs	CG Iterations	Initial Residual	Final Residual
339	71	4.074	3.13×10^{-10}
1,289	182	3.1133	2.78×10^{-10}
5,025	364	1.637	1.60×10^{-10}
19,841	DNC	0.869	9.6×10^{-7}
78,849	DNC	0.478	1.56×10^{-3}
314,369	DNC	0.277	9.92×10^{-2}

Conclusion

- We demonstrated the smoothing effects of Gauss Seidel on highly oscillatory components.
- We implemented a two-level method with GS smoothing and a coarse direct solve.
- We built a framework from which future multilevel test on NGSolve finite element spaces may be performed and tested it to demonstrate convergence.
- We implemented the method above as the preconditioner for a preconditioned conjugate gradient solver.
- After the work here, we implemented fully random coefficients using a Karhunen-Loève expansion. (Spoilers)

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